

Total No. of Questions—8]

[Total No. of Printed Pages—3

Seat No.	
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[5057]-253

S.E. (Computer Engineering) (First Semester)

EXAMINATION, 2016

DIGITAL ELECTRONICS AND LOGIC DESIGN

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—** (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
(ii) Figures to the right indicate full marks.
(iii) Assume suitable data, if necessary.

1. (a) Do the required conversions for the following numbers : [6]
(i) $(205.76)_{10} = ()_2$
(ii) $(7A2B)_{16} = ()_8$
(iii) $(6516)_{10} = ()_{16}$
(b) Define the following terms for TTL family : [2]
(i) Fan Out
(ii) Speed of Operation.
(c) Explain the operation of CMOS NOR gate. [4]

Or

2. (a) Minimize the following functions using K-map and realize using logic gates : [4]
$$F(A, B, C, D) = \sum m(0, 2, 8, 10, 14).$$

P.T.O.

- (b) Perform the following operation using 2's complement method : [2]

$$(27)_{10} - (14)_{10} = (?).$$

- (c) Explain the working of three input TTL NAND gate with Totem-pole output. [6]
3. (a) Implement the following function using 4 : 1 multiplexer : [4]
 $F(A, B, C, D) = \Sigma m(0, 3, 7, 9, 11, 14, 15).$
- (b) Convert the following Gray code numbers to Binary : [2]
(i) $(101101)_2$
(ii) $(111111)_2$
- (c) What are the applications of Flip-Flops ? Explain the working of SR Flip-Flop. [6]

Or

4. (a) Explain with suitable equations and diagrams concept of look ahead carry Generator for 4-bit adder circuit. [6]
(b) Design MOD 76 counter by using IC 7490. [6]
5. (a) Explain the following modelling styles of VHDL with suitable example : [6]
(i) Behaviour modelling style
(ii) Data flow modelling style.
- (b) What is ASM chart ? Explain components of ASM chart. What are applications of ASM chart in digital system design ? [7]

Or

6. (a) Draw an ASM chart and state table for 3-bit Up counter having control input E : [7]
- (i) If control input E = 0 : Counter remains in same state
- (ii) If control input E = 1 : Counter goes to next state.
- (b) What is VHDL ? Explain entity and architecture declaration in VHDL with suitable example. [6]
7. (a) Draw and explain the basic architecture of FPGA. [6]
- (b) A combinational circuits is defined by the functions : [7]
- $$F_1(A, B, C) = \Sigma m(0, 2, 5, 7)$$
- $$F_2(A, B, C) = \Sigma m(0, 1, 6, 7)$$
- Implement this circuit with PLA.

Or

8. (a) Comparison between PROM, PLA and PAL. [7]
- (b) What is CPLD ? Give the difference between CPLD and FPGA. [6]